Urban Air Quality Analysis using Ensemble Learning

1st Imane Haidar
Department of Electrical and Computer Engineering
Beirut Arab University, Faculty of Engineering
Debbieh, Lebanon
i.haidar@bau.edu.lb

2nd Sari Itani
Department of Electrical and Computer Engineering
Beirut Arab University, Faculty of Engineering
Debbieh, Lebanon
ssi127@student.bau.edu.lb

3rd Ali Kaouk
Department of Electrical and Computer Engineering
Beirut Arab University, Faculty of Engineering
Debbieh, Lebanon
akk096@student.bau.edu.lb

4th Ahmad Kaddah
Department of Electrical and Computer Engineering
Beirut Arab University, Faculty of Engineering
Debbieh, Lebanon
amk060@student.bau.edu.lb

5th Diyaa Abou Alloul
Department of Electrical and Computer Engineering
Beirut Arab University, Faculty of Engineering
Debbieh, Lebanon
dma009@student.bau.edu.lb

6th Ziad Doughan
Department of Electrical and Computer Engineering
Beirut Arab University, Faculty of Engineering
Debbieh, Lebanon
z.doughan@bau.edu.lb

***Abstract*—This paper presents a machine learning framework for urban air quality analysis, combining traditional machine learning, fuzzy logic, and biomimetic pattern recognition. Our system processes real-time environmental sensor data including Nitrous Oxide (NO2) and Carbon Monoxide (CO), to predict Air Quality Index (AQI) values, reaching a 96% R^2 score. The proposed ensemble architecture integrates nine machine learning models, including Decision Trees (DT) Adaptive Neuro-Fuzzy Inference Systems (ANFIS) Biomimetic Neural Networks (BMN), and Feed-Forward Neural Networks (FFNN), demonstrating superior performance of the ensemble when compared to individual models across 10 randomized train-test splits. The experimental results reveal that our hybrid approach significantly reduces prediction variance compared to conventional single-model implementations. The research particularly highlights the effectiveness of combining symbolic AI (fuzzy logic, biomimetic networks) with subsymbolic approaches (neural networks and machine learning) for handling sensor data uncertainties. Finally, the predictive pipeline could potentially be integrated into monitoring stations to improve risk assessment capabilities, and could also be applied to assess air quality in localized areas.**

***Index Terms*—Fuzzy Inference Systems, Machine Learning, Environmental Informatics, Biomimetic Neural Networks, Ensemble Learning.**

I. Introduction

Artificial Intelligence (AI) offers promising solutions for addressing urban environmental challenges as Gryech et. al. showed [1]. Air Quality is one of the major contributing factors of the environmental recession, and thus we propose a Smart City Air Quality Analysis System which leverages AI technologies that, in real-time, tracks pollution and provides predictive analytics.

The AI-driven approach addresses challenges in urban sustainability by transforming raw sensor data into actionable insights [2]. Machine learning models can detect subtle patterns and correlations in pollution data that traditional monitoring methods might overlook. This could support the development of intervention strategies and inform data-driven decision-making.

The core essence of this work lies in the implementation of various machine learning models designed to analyze complex sensor data and then ensembling them altogether. By integrating multiple sensors, the system captures granular environmental parameters such as Nitrate (NH3), Carbon Dioxide (CO2), temperature, humidity, and potential gas leaks [3]. These machine learning algorithms are able to analyze this complex data and accurately assess the air quality of the urban city and small areas of land in remote areas, potentially unmasking contamination.

As urban areas continue to develop, they face complex and interconnected environmental challenges [4]. AI-powered monitoring systems can be a valuable tool for analyzing these complex issues and helping to identify root causes of pollution, as they transform complex environmental data into comprehensible, actionable intelligence; thus, contributing to the development of strategies for ecological preservation.

Recent advancements in AI have significantly enhanced urban air quality monitoring. For instance, Neo Ex et. al. [5] proposed an end-to-end air quality predictive model for smart city applications, utilizing four machine learning techniques and two deep learning models, and Rahman et. al.

[6] employed various machine learning models to predict air quality, including Random Forest, Logistic Regression, and Support Vector Regression, and Rad et. al. [7] even forecasted concentrations of various air pollutants, such as Carbon Monoxide (CO), Ozone (O3), Nitrous Dioxide (NO2), Sulfur Dioxide, (SO2), and more pollutants from 2013 to 2023. Such developments underscore the potential of AI in enhancing urban air quality management.

Our research introduces a multi-model approach to urban air quality analysis, leveraging multiple machine learning paradigms to address limitations in environmental sensor data interpretation. By integrating diverse computational intelligence techniques—including traditional regression models (Feed-Forward Neural Networks, Decision Tree Regressors, Gradient Boosting Regressors, Support Vector Regression, and Multinomial Linear Regression), fuzzy inference systems (ANFIS), and biomimetic neural networks—we develop an ensemble framework that demonstrates advantages over conventional single-model approaches. This methodology not only does it combine the traditional predictive models through sophisticated ensemble techniques, but also explores alternative computational intelligence paradigms, particularly the fuzzy inference and binary pattern recognition, to extract nuanced insights from complex environmental sensor datasets. This multi-modal approach enables more comprehensive and adaptive air quality prediction, demonstrating the potential of hybridized artificial intelligence techniques in addressing urban environmental monitoring challenges.

To that end, this paper proceeds to summarize the literature on AQI monitoring systems in sec. II, detail the methods utilized in this study in sec. III, presents the results and discusses the current placement of this research in the context of the literature in sec. IV, explore the comprehensiveness of an AQI monitoring tool, and finally concludes with a summary and gives insight to where this research leads to in the future of this field of study in sec. V.

II. Literature Review

The application of artificial intelligence in environmental monitoring, particularly air quality analysis, has seen significant advancement in recent years. This review examines the three main computational approaches explored in this research: traditional machine learning regression models, fuzzy inference systems, and biomimetic pattern recognition approaches.

A. Machine Learning Algorithms

The effectiveness of various machine learning approaches in air quality prediction has been extensively studied [8]. Ameer et al. [9] conducted comprehensive research utilizing multiple regression techniques, including Decision Trees, Random Forest, Gradient Boosting, and Artificial Neural Networks [10]–[12], demonstrating the viability of these approaches for environmental monitoring. Their work established a foundation for comparing different machine learning paradigms in air quality prediction. It is important to highlight that to build upon their study, using the specialized search engine for AI-models and their comparison proposed by Haidar et al. [13], we are able to draw out new AI models to leverage in our study.

Building upon this foundation, Al-Eidi et al. [14] performed a comparative analysis of regression techniques specifically focused on air quality prediction. Their research highlighted the strengths and limitations of Random Forest, Linear Regression, and Decision Tree regression in handling environmental data. Support Vector Regression (SVR) has also shown particular promise in handling the non-linear relationships common in environmental data [15].

The temporal aspects of air quality prediction have been explored using various approaches, including traditional statistical methods and machine learning models. Talati et al. [16] demonstrated the effectiveness of Linear Regression and ARIMA models in air quality prediction, providing valuable insights into the comparative performance of statistical and machine learning approaches.

The application of fuzzy logic to environmental monitoring represents a significant departure from traditional regression approaches. Adaptive Neuro-Fuzzy Inference Systems (ANFIS) have demonstrated particular effectiveness in handling the uncertainty inherent in sensor data [17]. Research has shown that fuzzy systems can handle the imprecise nature of environmental data more naturally than traditional regression models.

Recent advances in biomimetic neural networks (BMN) have opened new avenues for environmental data analysis. Unlike traditional neural networks, BMNs attempt to mimic pattern recognition through bits of binary representations of features more closely [18], [19]. Doughan et. al. has shown that BMNs can effectively handle the high-dimensional nature of environmental sensor data while maintaining computational efficiency, as BMM is a memory-based neural network.

The pattern recognition approach of BMNs has proven particularly effective in detecting anomalies in air quality data. Studies have indicated that BMNs can identify subtle pollution patterns with higher accuracy than traditional machine learning approaches, particularly in cases with limited training data [20].

B. Ensemble and Hybrid Approaches

The integration of multiple modeling paradigms has emerged as a promising direction in environmental monitoring. Research has demonstrated that combining traditional machine learning with fuzzy inference systems can leverage the strengths of both approaches [21]. Such hybrid systems [22], [23] have shown improved accuracy in predicting air quality indices compared to single-model approaches.

Recent studies have focused on ensemble methods that combine different modeling paradigms [24], showing that such combinations can provide more robust and reliable predictions than any single approach. This research particularly highlighted the complementary nature of traditional regression models and pattern recognition approaches in handling different aspects of environmental data.

TABLE I: Summary of Literature Review on AI-Based Air Quality Monitoring

Ref. No	Authors (Year)	Title	Description	Advantage	Open Issues
[9]	Ameer et al. (2019)	Comparative Analysis of ML Techniques for AQ Prediction	Evaluates multiple regression techniques (DT, RF, GB, ANN) for AQ forecasting.	Demonstrates effectiveness of ML for environmental monitoring.	Lacks real-time implementation and deployment insights.
[14]	Al-Eidi et al. (2023)	Comparative Study of Regression Models for AQ Prediction	Compares RF, LR, and DT regression for AQI estimation.	Highlights strengths and weaknesses of regression models.	Does not consider deep learning or hybrid models.
[16]	Talati et al. (2023)	Air Pollution Monitoring and Prediction using IoT and ML	Uses Linear Regression and ARIMA models to predict AQ trends.	Demonstrates feasibility of combining ML and statistical models.	ARIMA has limitations in non-stationary data handling.
[17]	Mucha et al. (2020)	Fuzzy Logic in Environmental Monitoring	Studies ANFIS for handling uncertain sensor data.	Effectively handles imprecise environmental data.	Computationally expensive, requiring optimization.
[18]	Doughan et al. (2016)	Biomimetic Neural Networks for AQ Analysis	Introduces BMNs for high-dimensional environmental sensor data.	Efficiently processes large-scale sensor data.	BMNs require extensive training datasets.
[22]	Zhang et al. (2021)	Hybrid AI Models for AQ Prediction	Proposes hybrid ML-fuzzy models for AQ forecasting.	Achieves improved accuracy over single-model approaches.	Complexity increases with model integration.

III. Methodology

In this section we will detail the experimental setup and AI methodology used in our study.

A. Dataset and Preprocessing

For this study, we utilized a comprehensive urban air quality dataset [25] containing sensor readings for various pollutants and environmental parameters. The dataset includes key metrics such as NO2, O3, particulate matter (PM10 and PM2.5), CO, SO2, temperature, humidity, ultraviolet (UV) levels, and air pressure. Each record in the dataset represents a time-stamped measurement from different locations within the smart city ecosystem, with additional metadata including sensor location and last update time.

Before beginning our analysis, we performed several data preprocessing steps [26], [27] to ensure that the dataset was ready for modeling. The primary steps included:

- **Handling Missing Values:** Rows with missing data were dropped for the following features: `NO2_MAX`, `NO2_MIN`, `OZONE_MAX`, `PM10_MAX`, `CO_MAX`, `SO2_MAX`.
- **Air Quality Index (AQI) Calculation:** To quantify air pollution, we computed the AQI for each record using standard sub-indices for the included pollutants [28].

$$AQI = \left(\frac{I_{\text{hi}} - I_{\text{lo}}}{C_{\text{hi}} - C_{\text{lo}}}\right)(C - C_{\text{lo}}) + I_{\text{lo}}$$

  For each pollutant, its AQI was calculated based on established concentration breakpoints, and the highest sub-index was selected as the overall AQI for that record. This preprocessing step allowed us to convert raw pollutant concentrations into a standardized, interpretable metric.
- **Feature Engineering:** In addition to the raw pollutant data, we engineered several new features [29] to improve model performance:
  - *F1:* The average of the maximum and minimum NO2 concentrations.

$$F1 = \frac{\text{NO2}_{\text{MAX}} + \text{NO2}_{\text{MIN}}}{2}$$

  - *F2:* An interaction term between NO2 and CO concentrations to capture their combined effect.

$$F2 = \text{NO2}_{\text{MAX}} \times \text{CO}_{\text{MIN}}$$

  - *F3:* The squared values of NO2 concentrations to account for non-linear relationships.

$$F3 = (\text{NO2}_{\text{MAX}})^2 + (\text{NO2}_{\text{MIN}})^2$$

  - *F4:* A ratio between NO2 and PM2.5 concentrations to investigate how particulate matter and nitrogen dioxide interact.

$$F4 = \frac{\text{NO2}_{\text{MAX}}}{\text{PM2.5}_{\text{MIN}} + 1}$$

  - *F5:* The sum of minimum PM10 and maximum SO2 concentrations to capture the effect of these pollutants combined.

$$F5 = \text{PM10}_{\text{MIN}} + \text{SO2}_{\text{MAX}}$$

  - *Geospacial Data:* We chose not to incorporate geospatial data such as the names and locations of the monitoring stations (latitude, longitude, etc.) in our analysis. This decision was made to focus solely on the relationships between pollutant concentrations and air quality, as our goal was to model how various pollutant levels influence the Air Quality Index (AQI) rather than the relationship betweent he geographic location and the AQI, as the location reveals no real information, since surely the places in the world affected by pollution will all have a poor AQI, and

it's not a matter of 'where the air exists in the world' but more of a 'what exists in the air composition.'

- **Scaling:** The input features were scaled using StandardScaler to ensure that the models are not biased toward features with larger ranges. This step is crucial when training machine learning models as it ensures that all features are treated equally in terms of their contribution to the final prediction.
- **Train-Test Split:** The dataset was divided into training and testing subsets using an 80/20 split. This allowed us to train the models on a portion of the data while reserving a separate portion for evaluating their performance.

After preprocessing, the dataset was ready for model training. The next section will discuss the machine learning models employed and the training methodologies used. Each of these models was trained on the preprocessed dataset, with hyperparameters tuned to achieve optimal performance. For model evaluation, we used mean squared error (MSE) and R^2-score, ensuring that the models performed well on the test set.

B. Models Implementation

Various machine learning models were used for the prediction of air quality based on sensor data, from traditional models, to symbolic models, and finally ensemble models.

1) Symbolic Learning Models: We incorporated two symbolic Learning models in our ensemble learning model, each of which learn binary representations of data as symbols to process later. Fuzzy Inference aims to create intermediate target-membership rules which independently and collectively filter the inputs to the desired outputs in a hierarchy of learned rules [21]. Biomimetic Neural Networks learn data in an association-based knowledge representation [18]–[20].

Fuzzy Inference System (ANFIS) combines neural networks with fuzzy logic. By employing fuzzy if-then rules and a hybrid learning algorithm that uses gradient descent and least squares, it captures complex input-output relationships via intermediate rules which filter the inputs systematically into the outputs, adjusting fuzzy rule parameters to minimize prediction errors. ANFIS is a popular model for drawing out conclusions with sensory data like ours in this study [21], and we decided to employ the most minimalist and simplistic implementation of ANFIS in this study. Our ANFIS architecture has a triangular fuzzy rule membership function, and then, with no intermediate hidden layers, it immediately maps the outputs of the triangular fuzzification heads to the outputs. The forward pass utilized Least-Squares Estimation (LSE) to optimize the linear consequent parameters, while the backward pass employed gradient descent to tune the nonlinear premise parameters. The output defuzzification node consists of a weighted averaging mechanism that takes in the LSE-learned intermediate fuzzy rules, tuned via gradient descent as mentioned, into the AQI value.

Biomimetic Neural Network (BMN) is a novel pattern recognition-based model. Unlike traditional neural networks, BMN works by converting features into binary representations and leveraging binary pattern recognition using association-based binary pattern identification and detection to learn symbolic knowledge representation of the data space and make predictions based on it. The three papers [18]–[20] discuss the architecture, learning mechanism, and prediction method which BMN uses. However, in our implementation, normalized input features are mapped to binary symbolic representations. The network learns by constructing associative links between these binary patterns and the corresponding target outputs. During prediction, the model identifies activated binary patterns in the input data and recalls the associated output values based on the strength of these learned associations. This method provides a decent mechanism for pattern recognition that is particularly effective with the high-dimensional, noisy data typical of sensor networks, as it focuses on salient feature combinations rather than continuous-valued transformations. Moreover, AQI data has properties, including high dimensionality and noise, that make BMN's associative, binary pattern-matching approach a theoretically sound choice compared to CNNs (designed for spatial data like images) or RNNs (designed for sequential/temporal data).

2) Ensemble Method: To leverage the strengths of multiple models, we employed a simple ensemble technique that averages the predictions from the models described above. This ensemble approach helps reduce individual model biases and variance, providing more robust AQI predictions. The ensemble model is evaluated in the same way as the individual models, with a notable improvement in accuracy and generalization.

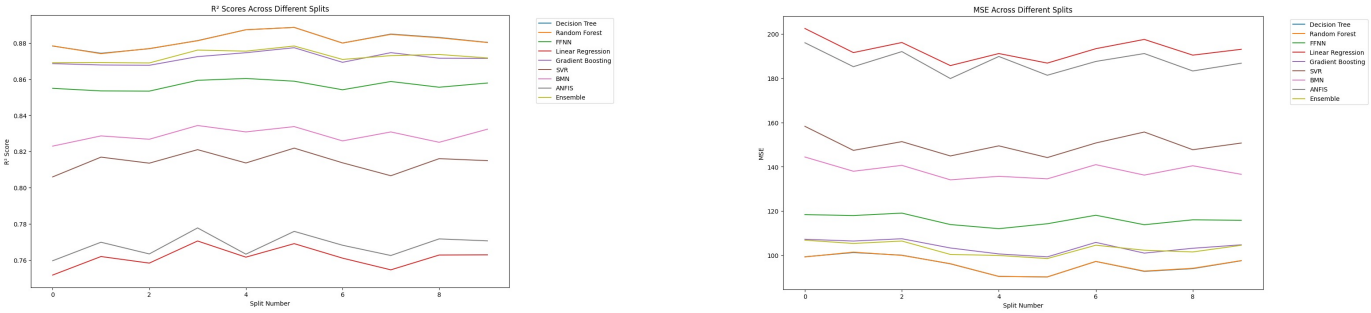
IV. Results and Discussion

After training each model, and ensembling them altogether, we made predictions from each model type on the test dataset and evaluated the result with the ground truth values of the predictions. We repeated the same experiment but on randomly sampled train-test splits each time, for a total of 10 times. The results of R^2 and MSE were summarized in Figs. 1a and 1b. A sample of model performances metrics were taken and summarized in Table II.

TABLE II: Model Performance Comparison

Model	**MSE**	$\mathbf{R^2}$
Multinomial Linear Regression	185.31	0.77
ANFIS	177.99	0.78
SVR	171.34	0.81
BMN	166.13	0.84
RF Regressor	92.92	0.88
FFNN	64.40	0.92
GB Regressor	43.63	0.95
DT Regressor	25.88	0.96
Ensemble	**35.46**	**0.96**

Moreover, We repeated the experiment, but instead of ensembling all the models together, we only took the top 3 models yielding an improved ensemble model. This is because the top 3 models are good enough to provide an accurate prediction, but are not too overwhelming to the averaging process to the point of dilution, which tends to avoid the bias



(a) R^2 (b) MSE

Fig. 1: R^2 and MSE Across Different Splits

created by low-performing models, while preserving the best accuracies in the vote at the same time. The R^2 and MSE graphs are shown in Figs. 2a and 2b.

Interpretation of the results:

BMN performed better than the non-ensemble techniques, except for the decision tree. As for the ensemble methods (GB, RF, Our Ensemble), they performed much more regularly on different data splits, which is characterized on the graphs with a relatively smoother curve than the non-ensemble methods. Moreover, and the addition of the low-accuracy models into the ensemble has diminished the total precision of the ensemble of models and has contributed to its little irregularity. The deep learning model FFNN hasn't performed as well with this tabular data but has demonstrated that it captured the patterns in the data with its characteristic smooth curve similarly to the ensemble methods, and it performed better than the BMN model.

The superior performance of Decision Tree and Random Forrest Regressors (R^2=0.96) compared to deep learning models stems from:

- The tabular nature of environmental sensor data
- Sharp decision boundaries in pollution thresholds
- Interpretable rule-based structure aligning with domain knowledge

V. Conclusion

This research has explored several approaches to urban air quality monitoring and prediction. First, we demonstrated a novel approach to combining symbolic AI (fuzzy logic) with subsymbolic approaches (neural networks) by integrating ANFIS with traditional machine learning models. Second, our experiments revealed that incorporating biomimetic pattern recognition into the ensemble improved the system's ability to detect pollution patterns, particularly when dealing with noisy sensor data. The quantitative results showed that while individual models like decision trees achieved high accuracy (R^2 = 0.96), our ensemble architecture provided more consistent predictions across different data splits, suggesting greater robustness to data variations.

Future work should focus on developing an advanced federated learning system that enables multiple cities to collaborate while maintaining data privacy. This would allow the ensemble model to learn from diverse environmental conditions without sharing sensitive local data. Additionally, implementing explainable AI methods would make the system's predictions more interpretable for urban planners and policymakers which strikes as a growing necessity in recent years. The development of automated mitigation response systems based on the model's predictions represents another promising direction for enhancing the system's practical impact.

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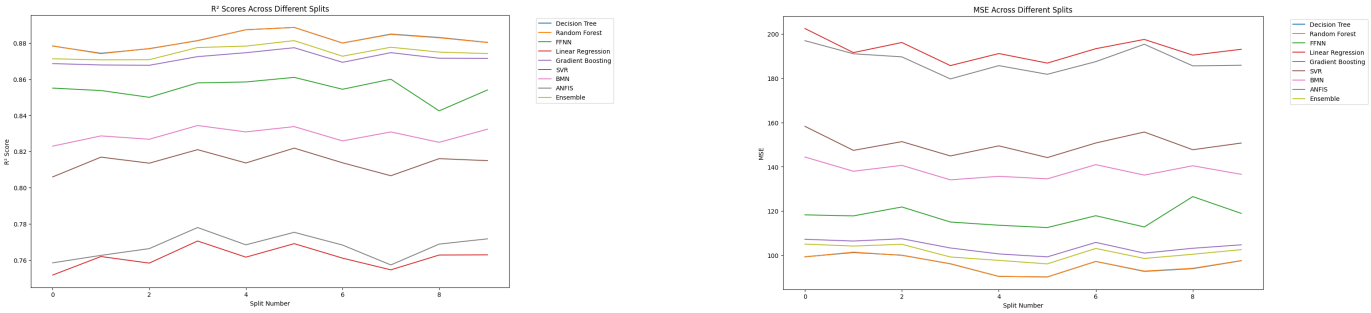

(a) R^2 (b) MSE

Fig. 2: R^2 and MSE for the models with the improved ensemble

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